

MCKELLAR-SIPES, TN, USA

WMO: 723346

Lat: 35.593N

Lon: 88.917W

Elev: 433

StdP: 14.47

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 03811

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	19.4	4.5	7.0	20.4	8.7	8.7	23.1	23.6	44.0	20.7	45.7	6.1	20	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	94.8	76.8	92.8	76.8	90.8	76.2	80.2	90.0	79.0	88.9	78.0	87.4	6.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.4	144.7	85.6	76.2	139.1	84.2	75.2	134.2	83.1	43.9	89.7	42.7	88.8	41.6	87.5	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.6	17.7	15.8	7.7	98.0	5.1	3.1	4.1	100.2	1.1	102.1	-1.8	103.8	-5.5	106.1
			6.7	82.0	5.0	1.4	3.1	82.9	0.2	83.7	-2.6	84.5	-6.2	85.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	60.4	39.3	43.1	51.2	60.8	69.3	76.8	79.4	78.7	72.0	60.9	49.7	42.2
	DBStd	16.63	11.87	11.05	10.31	8.80	7.04	4.81	4.09	4.39	6.38	8.54	9.60	10.43
	HDD50	1171	372	247	112	16	0	0	0	0	0	12	122	290
	HDD65	3444	796	614	439	180	42	1	0	0	17	183	464	708
	CDD50	4963	41	55	149	341	600	803	913	891	659	350	113	48
	CDD65	1762	1	2	11	55	177	354	448	426	226	56	5	1
	CDH74	17165	0	3	91	461	1534	3503	4585	4309	2109	540	29	1
	CDH80	7132	0	0	9	79	468	1479	2098	2026	845	126	2	0
Wind	WSAvg	6.5	8.3	8.3	8.3	8.2	6.7	5.4	4.7	4.3	4.5	5.6	6.7	7.3
Precipitation	PrecAvg	53.1	3.7	4.4	5.1	5.3	5.8	4.4	4.2	3.1	3.9	3.2	4.8	5.3
	PrecMax	76.0	8.7	10.3	12.3	11.3	14.4	11.1	11.7	10.2	13.1	7.4	10.3	12.5
	PrecMin	34.7	0.7	1.2	1.7	1.2	1.4	0.6	0.1	0.8	0.4	0.0	1.8	0.9
	PrecStd	9.9	2.1	2.4	2.7	2.6	3.0	2.3	2.5	2.0	2.7	2.0	2.2	3.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	70.3	73.7	81.4	85.5	90.2	94.5	97.2	99.6	95.1	87.6	77.9	72.9
		MCWB	62.8	64.2	65.0	68.9	73.6	75.8	78.3	76.5	73.5	70.6	65.5	65.0
	2%	DB	66.4	69.8	76.7	82.3	87.6	92.4	94.2	95.4	91.4	83.3	73.2	67.3
		MCWB	60.6	61.2	62.1	66.4	72.4	76.2	78.2	76.5	74.1	69.3	62.5	61.1
	5%	DB	62.9	65.9	72.6	79.5	85.1	90.4	92.2	92.6	88.4	80.3	69.9	63.3
		MCWB	57.6	57.5	60.3	65.0	70.9	75.5	77.9	76.8	73.0	66.5	60.8	57.9
	10%	DB	58.2	61.9	68.7	76.2	82.5	88.3	90.2	90.0	85.2	76.4	66.0	59.4
		MCWB	53.2	54.7	58.1	63.5	69.9	74.6	77.2	76.2	71.5	65.3	58.8	54.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	64.7	66.2	67.3	72.1	76.7	80.2	82.0	81.9	77.8	75.2	68.9	66.1
		MCDB	68.4	70.9	76.2	81.1	85.5	89.9	91.0	92.3	88.6	83.1	73.1	70.5
	2%	WB	62.1	63.7	64.7	69.4	74.9	78.7	80.4	80.0	76.3	72.7	65.8	62.5
		MCDB	65.8	68.6	73.6	78.2	83.9	88.7	90.1	89.8	86.7	79.5	70.4	66.1
	5%	WB	58.6	59.8	62.3	67.6	73.2	77.5	79.4	78.7	75.1	70.2	63.1	59.1
		MCDB	62.2	64.2	69.9	75.9	82.0	87.2	89.4	88.2	84.6	76.3	68.0	62.6
	10%	WB	53.9	55.7	59.9	65.7	71.5	76.2	78.2	77.5	73.6	67.6	59.8	55.0
		MCDB	57.5	60.3	67.4	73.2	80.0	85.1	87.6	86.8	81.5	73.7	64.8	58.5
Mean Daily Temperature Range	5% DB	MDBR	18.0	19.1	21.3	22.0	20.4	19.9	19.3	20.9	22.9	23.5	21.4	17.9
		MCDBR	20.4	23.4	25.0	24.5	22.3	22.2	21.5	24.1	26.0	26.7	25.2	20.6
		MCWBR	15.7	16.3	14.4	12.1	10.0	9.0	8.6	9.3	10.8	13.4	16.2	15.3
	5% WB	MCDBR	17.5	19.4	21.3	18.9	19.0	19.3	19.6	20.1	20.8	20.6	19.7	17.8
	MCWBR	15.1	15.8	14.3	10.8	9.6	8.9	8.6	9.3	9.6	11.9	15.3	15.1	
Clear-Sky Solar Irradiance	taub	0.315	0.325	0.364	0.390	0.430	0.446	0.486	0.492	0.420	0.363	0.329	0.314	
	taud	2.494	2.455	2.376	2.304	2.250	2.213	2.136	2.106	2.303	2.439	2.481	2.505	
	Ebn at Noon	281	291	287	284	272	267	255	251	265	273	273	274	
	Edh at Noon	26	30	36	41	44	45	49	49	38	31	26	25	
All-Sky Solar Radiation	RadAvg	751	914	1269	1638	1846	2041	1963	1798	1531	1193	844	635	
	RadStd	62	114	108	106	128	137	107	97	107	144	85	62	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page